

# Sound Insulation of Double Panels - Comparison of Existing Prediction Models

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## Summary

Numerous qualitative comparisons between existing prediction models for the sound insulation of double panels have been presented in the literature but quantitative comparison between models is still lacking. It is important to have impartial evidence of the applicability and accuracy of the prediction models before any of them can be applied for real double wall structures. The aim of this study was to present a qualitative and quantitative comparison of the existing double wall prediction models. Seventeen models were implemented. Quantitative comparison between models was made in four categories: without studs and cavity absorbent, without studs and with absorbent, rigid studs with absorbent, and flexible studs with absorbent. The quantitative comparison of the models was made against measurement data containing altogether 52 laboratory tests for double walls. The average prediction errors for individual frequencies were as high as 20 dB ( $R_w$ : 15 dB) for most of the models, while it was less than 10 dB ( $R_w$ : 5 dB) for the best models. None of the models was sufficiently accurate in all four categories. There is an obvious need to develop a new model, which can be reliably applied to arbitrary double and multilayer walls. In the present situation, the most accurate model should be selected according to the type of the double wall.

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## 1. Introduction

More than 20 prediction models have been found from international journals or well-known text-books. In this study, they are called original models (original papers). The exact number of models or their variations even might be in the hundreds if all international and national publications were included. In this study, only the most referred models are considered. So far, several qualitative comparisons between the original models have been presented in literature reviews but the quantitative comparison is lacking. This is a fundamental problem for a practitioner's point of view. It is tempting to believe that the original models presented in well-known text-books and journals are reliable, especially because successful comparisons with measured data are presented.

There are also several applied papers in which the accuracy of a certain prediction model has been studied. Such papers are important supplements to original papers because their authors are more independent than the authors of the original paper. The author of this article has also studied some models selected on the basis of convincing validation results presented either in the original paper or in applied papers [1, 2, 3, 4, 5, 6]. The main conclusion

of the author's previous studies was that none of the existing prediction models was able to take all important wall parameters into account. The original papers often give such an optimistic impression that the models should work with all types of double wall structures. The general limitations of the models are seldom discussed in the articles although the topics of the original articles are usually very general, dealing with prediction models for double walls. However, they usually apply to only some double wall types. A poor knowledge of building technology is also evident; the structures have no applications in real buildings. Because there is a general tendency that the authors of the original models over-emphasize the validity of their new model, a false impression can easily be formed. It is possible that the comparison prediction and measurements shown in original papers are not randomly chosen; it must be tempting for the authors to choose such a validation result where the best fit with the predicted and measured data is obtained. Statistically, the sample size has been insufficient in all previous studies to say anything in general about the prediction accuracy. In addition, some of the validated wall structures have been poorly documented so that the verification of the model by the readers is impossible.

On the basis of References [1]–[6], a strong desire arose to seriously compare the most referred double wall prediction models. There are several reasons for this. Firstly, it is important to make a distinction between good and poor

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models. Secondly, it is important to know the limits of each model because all models are not appropriate for all wall types. Thirdly, if a single and all-inclusive model cannot be selected, it is important to indicate the best properties of all existing models to be able to develop better models.

The aim of this study is to present a qualitative and quantitative comparison of the existing double wall prediction models. The validation of existing original models will be made with recent laboratory experiments in which the structures were well documented [7].

## 2. Literature review

The prediction models chosen for this study are presented in the Appendix. The original equations are repeated here to avoid any misunderstandings in their application. It was not the purpose of this study to modify the models.

The models have been presented either in the form of transmission coefficient,  $\tau$ , or sound reduction index,  $SRI$ . The sound transmission coefficient is defined as the ratio of transmitted,  $W_t$  (Nm), and incident sound power,  $W_i$ , by

$$\tau = W_t / W_i, \quad (1)$$

or as the ratio of squared sound pressures

$$\tau = \tau(\theta) = \left| \frac{\bar{p}_t(\theta)}{\bar{p}_i(\theta)} \right|^2, \quad (2)$$

where  $\theta$  is the sound incidence angle,  $p_i$  is the pressure of incident sound (Pa), and  $p_t$  is the pressure of transmitted sound. The sound reduction index is defined by

$$SRI = 10 \log (1/\tau). \quad (3)$$

The SRI depends on the sound incidence angle  $\theta$ . There are three types of prediction models regarding the sound incidence angle:

1. Normal incidence ( $\theta = 0^\circ$ ).
2. Field incidence without integration. The models are pre-corrected to result in random incidence ( $\theta = 0 \dots 90^\circ$ ) or field incidence ( $\theta = 0 \dots \theta_i$ ), where  $\theta_i = 70 \dots 85^\circ$  depending on the author.
3. Arbitrary angle. These models make the consideration of any range of angles possible. Transmission coefficient for field incidence is calculated using the Paris equation

$$\begin{aligned} \tau_d &= \frac{\int_{\theta=0}^{\theta_i} \tau(\theta) \sin(\theta) \cos(\theta) d\theta}{\int_{\theta=0}^{\theta_i} \sin(\theta) \cos(\theta) d\theta} \\ &= \int_{\theta=0}^{\theta_i} \tau(\theta) \sin(2\theta) d\theta. \end{aligned} \quad (4)$$

The limiting angle of sound incidence in reverberation chambers is usually  $\theta_i = 70 \dots 85^\circ$  (field incidence angle) to obtain good conformance between measurements and calculations. Slightly different limiting angles are used in the original paper and they are not changed in this study.

### 2.1. Double walls without studs

The first well-known model was introduced by Beranek and Work [8]. It was based on wave behaviour of sound. The propagation of sound in the cavity was based on the impedance approach, which enabled the modelling of multiple layers. Beranek and Work were interested only in normal sound incidence angles. This limits the general application of the model because all sound incidence angles exist in rooms. In addition, the bending stiffness of individual panels was not included in the model. For historical reasons, the model was included in this study.

London [9] presented a progressive-wave model, where the influence of the sound incidence angle was also considered making it applicable to normal rooms. The diffuse-field  $SRI$  was calculated from angle-dependent transmission coefficients using the Paris equation. Moreover, London introduced a real part to the panel impedance  $R$ , i.e. panel resistance as an addition to the mass reactance term. Panel resistance could not be physically measured but it was obtained by the best fit between the predicted and measured SRI for a single panel. However, the resistance term has not been used in newer models but it has been replaced by loss factor. London's model was restricted to double panels with an empty cavity. Neither asymmetric panels nor loss factors were considered. The models of Beranek and Work, and London are often referred to as classical models. They are based on the assumption that the wall is infinite in size.

In 1966, White and Powell [10] presented a mode-coupling model for resonant transmission of a finite rectangular panel. The statistical mechanical approach of Lyon and Maidanik [11] was applied to wall structures. The energy flow through a coupled multi-resonant system was considered as an analogue of a heat transmission problem. This model took the area of the wall into account, but neither cavity absorbents nor the cavity thickness was considered. The validity of this model for practical walls has been found questionable in several references. Therefore, it was not validated in this study.

London's model could not explain the behaviour of sound in the cavity for oblique sound incidence angles, nor the effect of absorbent in the cavity. Mulholland *et al.* [12] and Cummings and Mulholland [13] presented alternative models where ray tracing model was applied to take the lateral sound propagation in the cavity into account. In the first one, the absorption in the cavity was assumed to take place on the surfaces of the panels, and was described by the absorption coefficient of the panel ( $\alpha_p$ ). The bending stiffness of the panels was not considered. In the second one, absorption was assumed to be located on the edges of the cavity, and was described by the edge absorption coefficient ( $\alpha_c$ ). The latter approach was a great innovation because the sound field of the cavity does not only consist of resonances perpendicular to the panels. Typically, the thickness of the cavity of practical double walls is 50 mm...200 mm. Perpendicular resonances occur only at high frequencies when  $\lambda < d/2$ , that is, above 1000 Hz...3000 Hz. Thus, most of frequency band

is dominated by in-plane sound field in the cavity (low-order modes).

In 1970, Crocker and Price [14, 15] applied the statistical energy analysis for wall structures. The important progress after the model of White and Powell was that non-resonant vibration and cavity resonances were considered. Resonant and non-resonant transmission coefficients were dealt with separately. The angle-dependence of sound radiation was not considered but the model directly produced the response in the field incidence angles. The radiation resistance of the panel was calculated according to the theory of Maidanik [16]. This model included considerably more parameters than previous models and the equations were more complicated. The main problem of the model was the ignoring of the spring property of the air in the cavity. Therefore, the mass-air-mass resonance was ignored. The treatment of the cavity was a continuation of the model of Cummings and Mulholland [13]: the sound field of the cavity was divided into low-order modes (wavelength  $\lambda > d/2$ ) and high-order modes ( $\lambda < d/2$ ). Moreover, the size of the cavity was considered so that the division of the cavity into smaller units could be modelled. The cavity absorption was modelled using the absorption coefficient of the edges (low-order modes) or the panels (high-order modes).

In 1972, Donato [17] introduced a useful correction to take into account the size of the specimen. This correction could be applied to all models assuming infinite panel size. A user-friendly form of this correction was presented later by Elmallawany [18]. The correction was strongest at low frequencies where the dimensions of the test aperture were of the same order as the wavelength of sound in air. It is a typical feature of the classical prediction models that they underestimate the *SRI* at low frequencies compared to the measured *SRI*. Moreover, Donato facilitated the application of the classical models by modifying the integrals into a more analytical form. However, neither cavity absorbents nor asymmetry of panels were considered. It was not possible to reproduce the same *SRI* values as Donato above the critical frequency in this study despite of several trials and simple equations. That is, the verification failed at this frequency range. Therefore, Donato's model was ignored in the quantitative comparison.

Mulholland *et al.* [19] developed the impedance-transfer method of Beranek and Work for oblique incidence angles. However, only empty cavities were considered and no experimental results were presented to verify the theory. A similar extension was later presented by Ookura and Saito [20]. The pressure transmission ratio could be determined for each successive layer by its propagation factor and impedance. The model of Delany and Bazley [21, 22], was applied for the determination of the impedance and the propagation factor of porous materials from measured flow resistivity data. Unfortunately, only the impervious panel, the sound-absorbing cavity material, and the empty cavity were presented as elementary layers.

Heckl [23] used dynamic stiffness per unit area,  $s'$ , instead of impedance-related parameters, as the principal de-

scriptor of the cavity. It was assumed that the cavity is locally reacting. Therefore, this model is restricted to double panels having a highly sound-absorbing cavity.

Alternative progressive impedance methods have also been presented later. Hamada and Tachibana [24] based their approach on the utilization of the electronic four-terminal network theory. Each subsystem of the wall was described as a cascade matrix (F-matrix). The matrix method considerably facilitated the equations and the implementation of the algorithm, especially when multiple layers are studied. London's model was applied in the calculation of single panels. Fringuellino and Guglielmone [25] recently presented a model, which was very similar to that of Ookura and Saito, and that of Hamada and Tachibana.

Fahy presented two models for uncoupled double walls [26]. The first one applied to a double panel with arbitrary cavity absorption and to a normal sound incidence angle. Its applicability is very poor in building acoustics like the Beranek's model because the normal incidence angle gives a strong overestimation. No experiments were presented to verify the models. The models were modified in the new edition of the text-book [27]. The benefit of Fahy's models was that the lowest normal mode frequency,  $f_{11}$  (Hz), of the panel was considered. However, resonant transmission and loss factor were not considered. In this study, three models of Fahy are used, of which two are from [26] and the last is from [27]. The latest model also contains measurement data so that the verification of the model was possible.

Au and Byrne [28, 29] developed the impedance transfer method for predicting the insertion loss of lagging structures. The impedance for flexible and impervious layers, like rubber, and orthotropic profiled panels were introduced as new elementary layers. The disadvantage of their model was that it was designed to calculate the insertion loss of additional lightweight layers over existing walls or duct. An application for full double walls was presented later by Ver [30].

Kang *et al.* [31] presented an interesting supplement to Fahy's model. Actually, the supplement works also for all models that apply Paris' equation and enable the examination of empty cavities. The problem of Fahy's model was that the bandwidth of the mass-air-mass resonance frequency,  $f_{mam}$ , is too wide. Kang was able to confirm, both theoretically and experimentally by intensity measurements, that the distribution of sound incidence angles would not be considered correctly by the Paris' equation. The Paris' equation should be weighted by a Gaussian distribution function so that the angles close to normal incidence have stronger weight than angles close to grazing incidence. The weighting has a strong effect when the cavity is empty. When the cavity is absorbing, the grazing sound incidence angles are effectively attenuated so that the Gaussian weighting is not necessary. This feature should be considered in all models when empty cavities are involved.

## 2.2. Double walls with studs

The above-mentioned models ignored the effect of studs. They can be applied only when both leaves of wall have separate studs but no common rails. This omission essentially restricts their application to real walls because studs connecting the panels together, or at least common rails in the ceiling or floor, are used in most practical double wall structures.

The influence of sound bridges on the sound insulation of double walls was studied by Sharp [32, 33]. He presented a simple “spreadsheet” model to predict the *SRI* of double walls with sound bridges. The transmission of the sound between the panels was divided into two paths: the airborne path (cavity coupling) and sound bridges (structural coupling). It was assumed that the cavity was highly sound-absorbing, that is, free from all kinds of resonances. Firstly, he derived simple analytic equations for the *SRI* of an uncoupled double panel for diffuse airborne excitation, including sound incidence angles  $0^\circ \dots 78^\circ$ . The classical model of London for double panels was adopted, but the *SRI* of individual panels was approximated by the simplified field-incidence model of Cremer [34]. The integration using Paris’ equation was replaced by +5 dB correction for single panels so that good agreement with field incidence was obtained, and numerical integration could be avoided. For rigid sound bridges between panels in a double wall, he applied the rigid point and line impedances introduced by Cremer and Heckl [35]. The only new parameter to be defined was the distance between rigid sound bridges,  $b$ .

Sharp found that for ideal uncoupled double walls with absorbing cavity, the addition of *SRI* per octave was 6 dB below the mass-air-mass resonance frequency (mass law) and 18 dB above it. Above a certain limit frequency, the slope decreased to 12 dB per octave. Above the critical frequency of panels, the slope was 18 dB per octave. For double walls with sound bridges, the previous shape was followed until the bridge frequency which is usually situated around 200 Hz. Above this, only an addition of 6 dB per octave can be achieved. That is, the slope is the same as for mass law. The surpass over the mass-law curve defined by the total mass of the wall increases with a decreasing number of studs. The disadvantages of Sharp’s model were that it contained several approximations and presumptions. In addition, the model for sound bridges was valid only below the lowest critical frequency of the panels.

Later, Gu and Wang [36] developed Sharp’s model for flexible sound bridges, which are usually made of steel. It was found that the addition of *SRI* was steeper, 12 dB per octave, above the bridge frequency. This model presupposes knowledge of the dynamic stiffness of the studs. Fahy [26] also presented an additional model to consider rigid sound bridges separated by a distance  $b$ . It can be applied separately to all three models of Fahy.

The weak point in the models of Sharp, and Gu and Wang, was that the absorption of the cavity boundaries was assumed to be perfect. This shortcoming was partially rectified by Davy [37]. He developed a hybrid model using

the ideas of several authors mentioned in this section. The absorption coefficient of the cavity was used to describe the losses in the cavity. His model also considers rigid and flexible studs. Another weak point in Sharp’s model was that the transmission through sound bridges above the critical frequency of the panels could not be modelled. This point was also rectified by Davy [38]. Davy’s model was also adopted in the handbook of Bies and Hansen [39]. Davy also studied the behaviour of the *SRI* at the mass-air-mass resonance for oblique sound incidence angles  $\theta$ . It was found that the shape of the *SRI* curve around the mass-air-mass resonance was flat for field sound incidence because the resonance frequency does not occur at a single frequency but at many frequencies simultaneously; its value increases with increasing sound incidence angle. This holds especially for an empty cavity. The limiting angle of sound incidence,  $\theta_l$ , was suggested to depend on the frequency and the area of the panel using the theory of Sewell [40] instead of a fixed limiting angle.

For further analysis of this study, 17 original models were selected. The qualitative comparison of the models is presented in Tables I and II. Table I gives a rough description of the models, how they deal with studs, cavities, symmetry of panels, stiffness of the panels and sound incidence angle distribution. For most of the models, verification data was available which means that it was possible to check whether the implementation of the model in this study was the same as in the original study.

The input variables of the 17 selected models are listed in Table II. The amount of input data varies strongly between the models. The number of calculation parameters ranges from 4 to 14. The models of Davy, and Price and Crocker seemed to be the most versatile ones.

## 3. Materials and Methods

The practical calculation work was done using SRICALC software, into which all models were programmed. The program has been under continuous development and revision in the laboratory since 1997, and it has been applied in several product development projects presupposing high accuracy.

The *SRI* results are presented in 1/3-octave bands in the range 50 Hz... 10000 Hz. However, the calculations in each third-octave band were performed in 1/9-octaves with field and normal sound incidence models. For models using Paris’ equation, the calculations were made using a resolution of 1/63-octave. The integration of the sound incidence angle from  $0^\circ$  to  $\theta_l$  was implemented using numerical integration. This was usually done using a resolution of  $d\theta = 0.1^\circ$ .

The main scientific work of this study consists of verification and validation of the models. Verification means that the prediction result of SRICALC is compared against the prediction result presented in the original paper. The structure presented in the original paper was used whenever possible. Verification was done for each model to check that the equations in SRICALC were implemented

Table I. Qualitative comparison of the prediction models. Abbreviations: A: Symmetric panels required; B: Critical frequency considered; C: Verification results not available;  $\theta$ : Sound incidence angle.

| Model |                          | year     | Stud types | Absorption modelling used | range of $\theta$ |               |
|-------|--------------------------|----------|------------|---------------------------|-------------------|---------------|
| 1     | Beraneck and Work        | [8]      | 1949       | -                         | $\Gamma_c$        | normal        |
| 2     | London                   | [9]      | 1950       | -                         | empty             | arbitrary A,B |
| 3     | Mulholland <i>et al.</i> | [12]     | 1967       | -                         | $\alpha$          | arbitrary A   |
| 4     | Cummings and Mulholland  | [13]     | 1968       | -                         | $\alpha$          | arbitrary     |
| 5     | Price and Crocker        | [15]     | 1970       | -                         | $\alpha$          | random B      |
| 6     | Donato                   | [16]     | 1972       | -                         | -                 | arbitrary A,B |
| 7     | Sharp                    | [32, 33] | 1978       | rigid                     | absorbing         | field B       |
| 8     | Ookura and Saito         | [20]     | 1978       | -                         | $Z_c$             | arbitrary B   |
| 9     | Heckl                    | [23]     | 1981       | -                         | $s'$              | arbitrary B   |
| 10    | Gu and Wang              | [36]     | 1983       | flexible                  | absorbing         | field B       |
| 11    | Fahy                     | [26]     | 1985       | rigid                     | $\Gamma_c$        | normal C      |
| 12    | Fahy                     | [26]     | 1985       | rigid                     | -                 | arbitrary C   |
| 13    | Hamada and Tachibana     | [24]     | 1985       | -                         | $Z_c, \Gamma_c$   | arbitrary B   |
| 14    | Au and Byrne             | [28, 29] | 1987       | -                         | $Z_c, \Gamma_c$   | arbitrary B   |
| 15    | Davy                     | [44]     | 1991       | both                      | $\alpha$          | random B      |
| 16    | Kang                     | [31]     | 2000       | -                         | -                 | arbitrary     |
| 17    | Fahy                     | [27]     | 2001       | -                         | $Z_c$             | normal        |

Table II. Physical parameters used in the prediction models (model numbers according to Table I and to the Appendix section numbers).  $N$  is the total number of input parameters. For abbreviations of double wall parameters, see Appendix.

| Model | panels |     |     |        |       |     |           |           | cavity |     |            |            |      |           |           |     | studs |     | environment |       |       |       |          | $N$ |         |   |
|-------|--------|-----|-----|--------|-------|-----|-----------|-----------|--------|-----|------------|------------|------|-----------|-----------|-----|-------|-----|-------------|-------|-------|-------|----------|-----|---------|---|
|       | $m'$   | $E$ | $R$ | $\eta$ | $\nu$ | $h$ | $L_{a,p}$ | $L_{b,p}$ | $d$    | $Z$ | $\alpha_p$ | $\alpha_c$ | $s'$ | $L_{a,c}$ | $L_{b,c}$ | $s$ | $Y$   | $b$ | $\kappa'$   | $V_1$ | $V_2$ | $T_2$ | $\theta$ |     | $\beta$ |   |
| 1     | x      |     |     |        |       | x   |           |           | x      | x   |            |            |      |           |           |     |       |     |             |       |       |       |          |     | 4       |   |
| 2     | x      | x   | x   |        |       | x   | x         |           | x      |     |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | 7       |   |
| 3     | x      |     |     |        |       | x   |           |           | x      |     | x          |            |      |           |           |     |       |     |             |       |       |       |          | x   | 5       |   |
| 4     | x      |     |     |        |       | x   |           |           | x      |     |            | x          |      | x         |           | x   |       |     |             |       |       |       |          | x   | 7       |   |
| 5     | x      | x   |     |        | x     | x   | x         | x         | x      |     |            | x          |      | x         |           | x   |       |     |             |       | x     | x     | x        |     | 14      |   |
| 6     | x      | x   |     |        | x     | x   | x         | x         | x      |     |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | 9       |   |
| 7     | x      | x   |     |        | x     | x   | x         |           | x      |     |            |            |      |           |           |     |       | x   |             |       |       |       |          |     | 7       |   |
| 8     | x      | x   |     |        | x     | x   | x         |           | x      | x   |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | 8       |   |
| 9     | x      | x   |     |        | x     | x   | x         |           | x      |     |            |            | x    |           |           |     |       |     |             |       |       |       |          | x   | 8       |   |
| 10    | x      | x   |     |        | x     | x   | x         |           | x      |     |            |            |      |           |           |     |       | x   | x           |       |       |       |          |     | 8       |   |
| 11    | x      |     |     |        | x     |     |           | x         | x      | x   |            |            |      |           |           |     |       | x   |             |       |       |       |          |     | 8       |   |
| 12    | x      |     |     |        | x     |     | x         | x         | x      |     |            |            |      |           |           |     |       | x   |             |       |       |       |          | x   | 8       |   |
| 13    | x      | x   | x   |        |       | x   | x         |           | x      | x   |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | 7       |   |
| 14    | x      | x   |     |        |       | x   | x         |           | x      | x   |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | 7       |   |
| 15    | x      | x   |     |        | x     | x   | x         | x         | x      |     |            | x          |      |           |           |     |       | x   | x           |       |       |       |          | x   | 12      |   |
| 16    | x      | x   |     |        | x     | x   | x         |           | x      |     |            |            |      |           |           |     |       |     |             |       |       |       |          | x   | x       | 8 |
| 17    | x      |     |     |        | x     |     | x         | x         | x      | x   |            |            |      |           |           |     | x     | x   |             |       |       |       |          |     | 9       |   |

in a similar manner as in the original study. Validation is meaningful only when verification results are acceptable. Verification error is the difference between the SRICALC prediction and the prediction in the original paper. The verification of a model was made by selecting 1–2 prediction results from the original paper and comparing the results obtained by SRICALC with each other. To be able to perform a correct verification, it was presupposed that all material information was given in the paper. In many cases, some input data were missing. Usually, the missing input data could be deduced from pictures or other material databases and verification could be done reasonably well.

The validation of the models is the main point of this study because it gives the accuracy of the models and we can compare the models with each other. Validation means that the prediction result of SRICALC is compared with the measured  $SRI$  of a carefully documented wall structure, model by model. The validation of models was done against measurement data obtained from the previous study where the author was also involved [7]. The study presented a parametric study of double panels (52 sound insulation tests) in which several cavity parameters were changed. The double panels were tested in an aperture of size 2.5 m<sup>2</sup> (1.1 m to 2.2 m).

Table III. The material data and literature references used in the verification of models 1–17. The middle column gives the Figure where the data was presented in the original paper. For abbreviations of double wall parameters, see Appendix. \*: Data given in [12, Figure 2]; \*\*: Data given in [41, Figures 8-9]. The word “missing” implies that the data value was not given in the article but its value could be estimated either by using given material information (e.g. aluminium) or by trial and error (best fit).

| Model | cavity                | Figure | Material data given in the article                                                                                                                                                                                                                       |
|-------|-----------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | empty                 | *      | $m'_1 = m'_2 = 5 \text{ kg/m}^2$ , $h_1 = h_2 = 1.86 \text{ mm}$ , $d = 0.305 \text{ m}$                                                                                                                                                                 |
| 1     | absorbing             | 6      | $m'_1 = 1.4 \text{ kg/m}^2$ , $m'_2 = 0.3 \text{ kg/m}^2$ , $d = 0.05 \text{ m}$ , $r = 1000 \text{ rayl/m}$ (Structure IV)                                                                                                                              |
| 2     | empty                 | 4      | $m'_1 = m'_2 = 1.2 \text{ kg/m}^2$ , $E_1$ , $E_2$ missing (aluminium), $R_1 = R_2 = 2.16$ , $h = 1/64''$ , $d = 3''$ , $\theta_i = 90^\circ$                                                                                                            |
| 3     | empty                 | 6      | $m'_1 = m'_2 = 5 \text{ kg/m}^2$ , $d = 0.305 \text{ m}$ , $\alpha_p = 0$ , $\theta_i = 80^\circ$                                                                                                                                                        |
| 3     | absorbing             | 6      | $m'_1 = m'_2 = 5 \text{ kg/m}^2$ , $d = 0.305 \text{ m}$ , $\alpha_p = 1$ , $\theta_i = 80^\circ$                                                                                                                                                        |
| 4     | absorbing             | 5      | $m'_1 = m'_2 = 0.47 \text{ kg/m}^2$ , $d = 0.0711 \text{ m}$ , $\alpha_c = 1$ , $L_{a,c} = L_{b,c} = 2 \text{ m}$ (missing), $\theta_i = 80^\circ$                                                                                                       |
| 5     | absorbing             | 6      | $m'_1 = 17.1 \text{ kg/m}^2$ , $m'_2 = 8.5 \text{ kg/m}^2$ , $d = 0.071 \text{ m}$ , $\alpha_c$ was tabulated, $L_{a,c} = 1.97 \text{ m}$ , $L_{b,c} = 1.55 \text{ m}$ , $\eta_1 = \eta_2 = 0.005$ (missing), $E_1$ and $E_2$ missing (aluminium)        |
| 6     | empty                 | 4d     | $m'_1 = m'_2 = 22.45 \text{ kg/m}^2$ , $E_1$ , $E_2$ missing (plasterboard), $\eta_1 = \eta_2 = 0.015$ , $d = 9.2 \text{ cm}$ , $L_{a,p} = 2.43 \text{ m}$ , $L_{b,p} = 3.05 \text{ m}$ , $\theta_i = 78.5$ (missing), area correction made              |
| 7     | absorbing and studded | 12     | $m'_1 = 14 \text{ kg/m}^2$ (missing), $m'_2 = 9 \text{ kg/m}^2$ (missing), $E_1$ , $E_2$ missing (plasterboard), $h_1 = 15.9 \text{ mm}$ , $h_2 = 9.5 \text{ mm}$ , $d = 0.102 \text{ m}$ , $b = 0.61 \text{ m}$                                         |
| 8     | empty                 | 6      | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (aluminium), $h_1 = h_2 = 2 \text{ mm}$ , $d = 50 \text{ mm}$ , $\theta_i = 85^\circ$ (missing), Donato correction not applied                                                                                   |
| 9     | absorbing             | 17a    | $m'_1 = m'_2 = 12.5 \text{ kg/m}^2$ , $E = 5 \cdot 10^9 \text{ Pa}$ missing (plasterboard), $\eta = 0.01$ , $d = 0.05 \text{ m}$ , $r = 28000 \text{ Pa s/m}^2$ , $\theta_i = 78^\circ$                                                                  |
| 10    | absorbing             | 8c     | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (plasterboard) $h_1 = h_2 = 12 \text{ mm}$ , $d = 0.07 \text{ m}$ , $b = 0.6 \text{ m}$ (missing), $K_B = 9 \cdot 10^6 \text{ N/m}$                                                                              |
| 13    | empty                 | 5      | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (cement), $h_1 = h_2 = 3 \text{ mm}$ , $R = 2.16$ , $S = 4 \text{ m}^2$ , $d = 0.05 \text{ m}$ , $\theta_i = 90^\circ$                                                                                           |
| 13    | absorbing             | 5      | as above but $r = 10000 \text{ Pa s/m}^2$                                                                                                                                                                                                                |
| 14    | empty                 | **     | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (steel), $h_1 = h_2 = 1 \text{ mm}$ , $\eta = 0.01$ , $d = 50 \text{ mm}$ , $\theta_i = 78^\circ$                                                                                                                |
| 14    | absorbing             | **     | as above but $r = 3000 \text{ Pa s/m}^2$                                                                                                                                                                                                                 |
| 15    | absorbing and studded | 3      | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (plasterboard), $h_1 = h_2 = 13 \text{ mm}$ , $d = 92 \text{ mm}$ , $\alpha_c = 1$ (missing), $h_1 = h_2 = 0.006$ (missing), $b = 0.6 \text{ m}$ , $C_m = 10^{-6} \text{ m/N}$ , $\theta_i = 78^\circ$ (missing) |
| 16    | empty                 | 11     | $m'_1$ , $m'_2$ , $E_1$ , $E_2$ missing (steel), $h_1 = h_2 = 2.2 \text{ mm}$ , $d = 6 \text{ mm}$ , $\beta = 2$ , $\theta_i = 90^\circ$                                                                                                                 |
| 17    | absorbing             | 11.12  | $m'_1 = m'_2 = 3 \text{ kg/m}^2$ , $d = 0.2 \text{ m}$ , $s = h = K = 1$ , $r = 0 \text{ Pa s/m}^2$                                                                                                                                                      |
| 17    | empty                 | 11.12  | as above but $r = 10000 \text{ Pa s/m}^2$                                                                                                                                                                                                                |

The prediction error,  $E$ , is defined as the difference between the predicted and measured  $SRI$  at a specified frequency  $f$ :

$$E(f) = SRI_{\text{pred}}(f) - SRI_{\text{meas}}(f). \quad (5)$$

The average prediction error of  $R_w$  for  $n$  experiments used in the validation process was calculated by

$$\overline{E}(R_w) = \frac{1}{n} \sum_{i=1}^n R_{w,i,\text{pred}} - R_{w,i,\text{meas}}. \quad (6)$$

This gives the trend of the prediction model (the sign is included). The standard deviation gives the number within which the prediction error is found with a 68% probability. The error of  $R_w$  can give a slightly false impression of the accuracy, because the  $R_w$  of lightweight double panels is mostly based on the performance at low frequencies. That is, the performance of the model at low frequencies will be overemphasized. Therefore, another single number descriptor was used together with  $\overline{E}(R_w)$ , which considers the errors at individual frequencies. The prediction error,  $E(AVE)$ , is the average prediction error over the frequencies 100 Hz...3150 Hz (16 bands). Thus, the average

prediction error over  $n$  experiments used in the validation process is

$$\overline{E}(AVE) = \frac{1}{n} \sum_{i=1}^{16} SRI_{i,\text{pred}} - SRI_{i,\text{meas}}. \quad (7)$$

The basic properties of the models, described in Tables I and II, differ considerably. Therefore, the prediction models were divided into four categories, within which the comparison was meaningful:

1. without studs and cavity absorbent ( $n = 5$ ),
2. without studs and with cavity absorbent ( $n = 16$ ),
3. rigid studs and with cavity absorbent ( $n = 25$ ),
4. flexible studs and with cavity absorbent ( $n = 6$ ).

The number  $n$  in parenthesis gives the number of measured structures against which the prediction models were tested (altogether 52 specimens).

## 4. Results

### 4.1. Verification of programming

The structures used in the verification are given in Table III. The structures are the same as those used in the

Table IV. Statistics of the verification errors of Figure 1 (dB).  $\sigma_E$ : Standard deviation of  $E(AVE)$ .

| Model |                  | $E(R_w)$ | $E(AVE)$ | $\sigma_E$ |
|-------|------------------|----------|----------|------------|
| 1     | empty cavity     | 2        | 0.5      | 2.7        |
| 1     | absorbing cavity | 5        | 1.9      | 4.9        |
| 2     |                  | 0        | 0.0      | 0.5        |
| 3     | empty cavity     | 1        | 0.8      | 1.1        |
| 3     | absorbing cavity | 0        | -0.5     | 0.5        |
| 4     | absorbing cavity | 0        | 0.1      | 0.3        |
| 5     |                  | 1        | 1.4      | 1.6        |
| 6     |                  | -5       | -0.9     | 2.5        |
| 7     |                  | -1       | -0.3     | 0.5        |
| 8     |                  | 0        | 0.3      | 1.6        |
| 9     |                  | -2       | 1.5      | 3.0        |
| 10    |                  | 0        | -0.7     | 0.6        |
| 13    | empty cavity     | -2       | -0.5     | 3.5        |
| 13    | absorbing cavity | 2        | 3.8      | 2.7        |
| 14    | absorbing cavity | -4       | -2.9     | 2.2        |
| 14    | empty cavity     | -2       | -1.2     | 1.4        |
| 15    |                  | -2       | 1.4      | 0.7        |
| 16    |                  | 0        | -0.2     | 1.2        |
| 17    | absorbing cavity | 1        | 0.9      | 0.6        |
| 17    | empty cavity     | 1        | 1.2      | 1.7        |

original papers. The verification errors are presented in Figure 1. Positive error means that SRICALC resulted in a higher value than the original paper. The statistics of Figure 1 are presented in Table IV. It should be noted that verification error does not express the accuracy of the model but the success of the programming.

#### 4.2. Validation of the models

Examples of individual validation results for all four categories are presented in Figures 2–5. The statistical comparison is presented in Figures 6–9. It shows the signed average prediction errors for  $E(R_w)$  and  $E(AVE)$  and their standard deviations. If the prediction error is positive, the model systematically overestimates the  $SRI$ . The higher the standard deviation, the less stable the prediction error.

### 5. Discussion

#### 5.1. Verification of programming

A perfect verification result would be that the curves in Figure 1 do not deviate from zero. However, this could not be achieved with any model (Table IV). A small and systematic error source was included in the method of obtaining the original  $SRI$  values from the original papers. The only available method was reading the  $SRI$  values from enlarged figures presented in the original papers of Table III. The estimated reading error was  $\pm 1$  dB... 2 dB. However, the reading error should not lead to any systematic positive or negative trend in the average error, just an increase in standard deviation.

The most obvious reason for the verification error was improper documentation of structures in the original papers, which actually prevents an appropriate verification

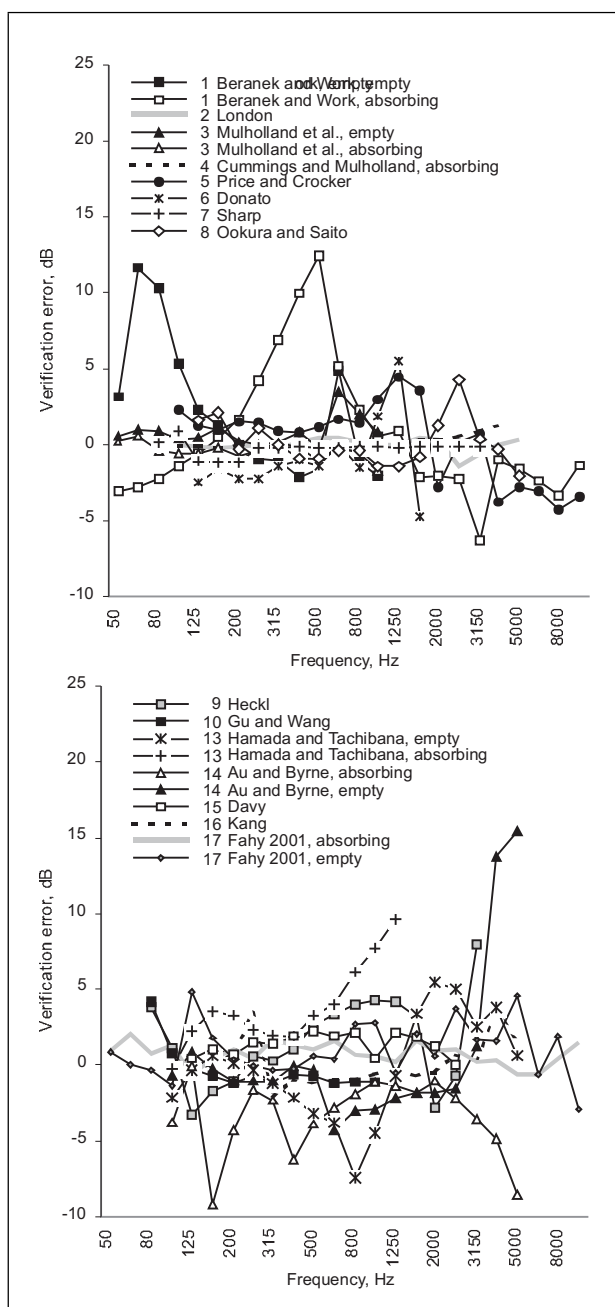


Figure 1. Verification results of the 17 prediction models. The verification error expresses the difference between the  $SRI$  according to SRICALC and the original paper.

work. For example, the error in Sharp's model is caused by the fact that the surface mass of the panels was not given at all. The surface mass of typical Finnish plaster-board was used, resulting, fortunately, in a good result. The same procedure was followed with Davy's model, where even more parameters were missing. There were similar problems also with other models. This is easily observed in Table III where word "missing" is used whenever data was not given. It is very probable that minor or major calculation errors could remain also in the original papers because the calculations are not checked before their publication by any other party than the authors.

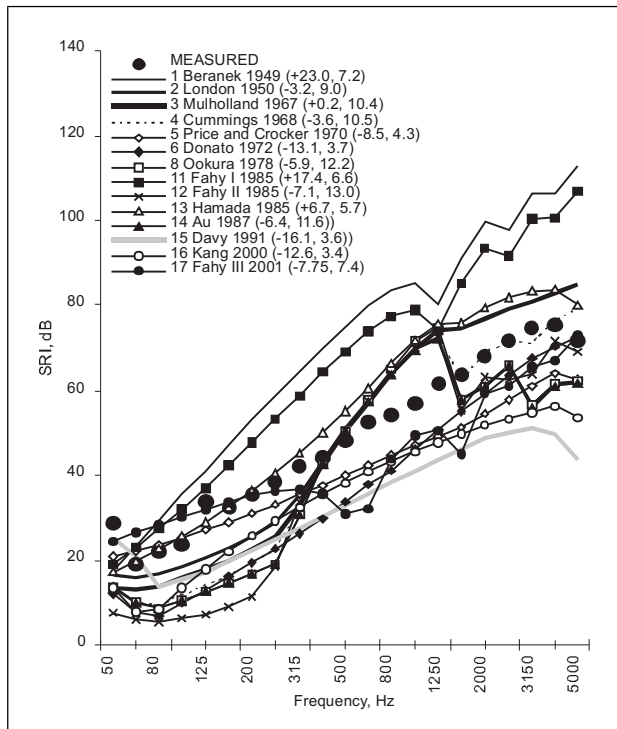


Figure 2. Example of validation in category 1 (without studs and cavity absorbent). The numbers in parentheses are the average prediction error  $E_{AVE}$  and its standard deviation in the frequency range 100 Hz... 3150 Hz.

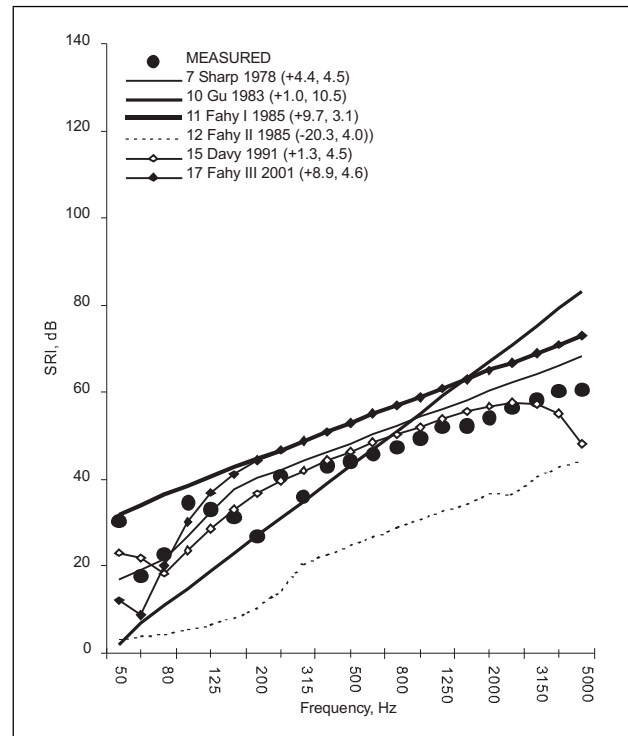


Figure 4. Example of validation in category 3 (rigid studs and with cavity absorbent).

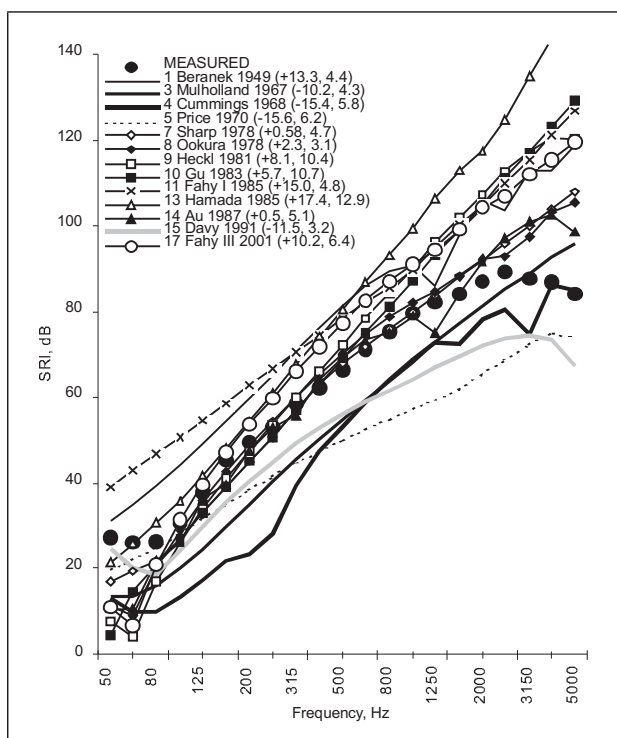


Figure 3. Example of validation in category 2 (without studs and with cavity absorbent).

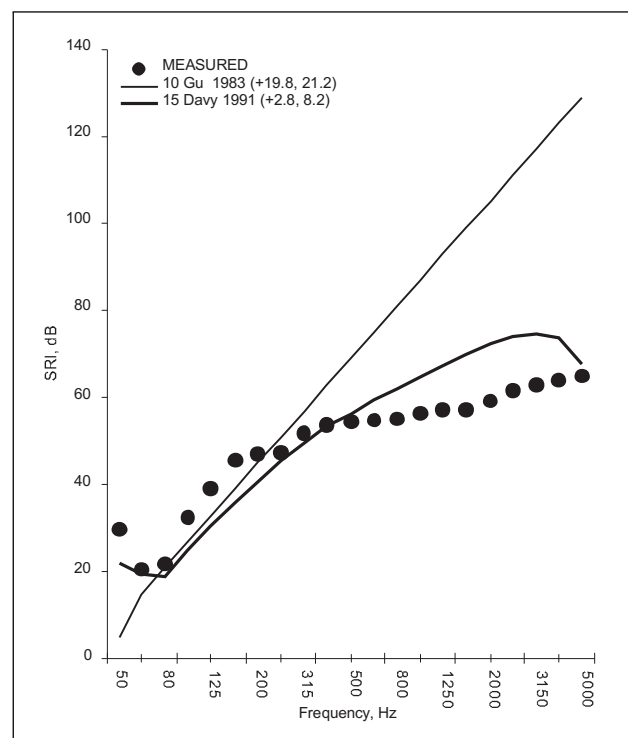


Figure 5. Example of validation in category 4 (rigid studs and with cavity absorbent).  $K_B = 2 \cdot 10^5$  N/m.

In several studies, the range of sound incidence angles used in equation (4) was not reported. It has a strong influence on the  $SRI$  of the single panel. When not docu-

mented, the correct value of  $\theta_l$  was chosen according to the best fit with the predicted  $SRI$  in the original paper. The verification errors fluctuated for some models, i.e. high



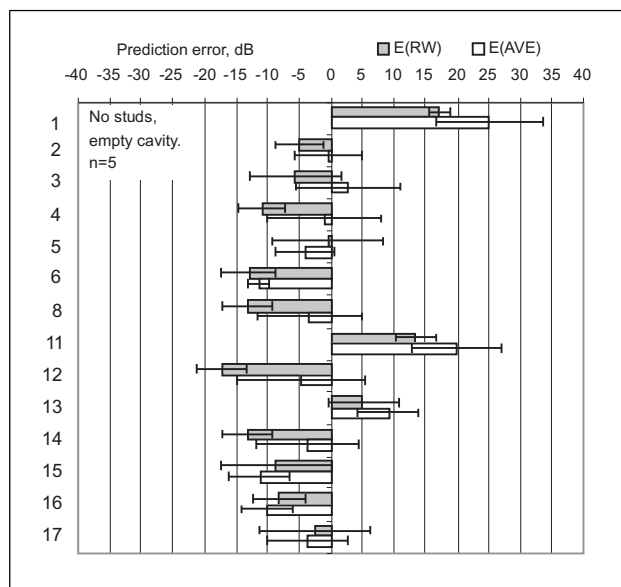


Figure 6. Average prediction error and standard deviation of  $E(R_w)$  and  $E(AVE)$  in category 1.

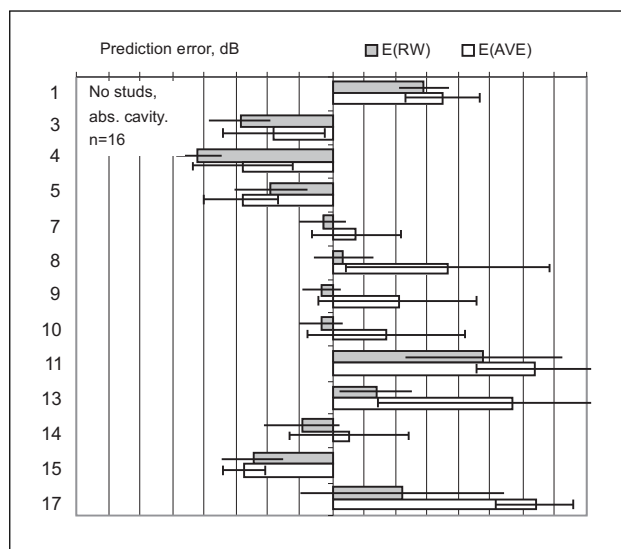


Figure 7. Average prediction error and standard deviation of  $E(R_w)$  and  $E(AVE)$  in category 2.

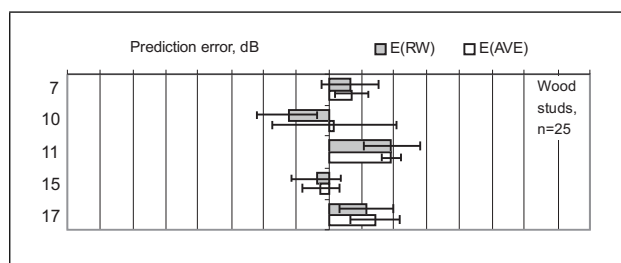


Figure 8. Average prediction error and standard deviation of  $E(R_w)$  and  $E(AVE)$  in category 3.

standard deviation but reasonably low average error. Such models were, e.g. those of Donato, Price and Crocker, Heckl, Au and Byrne, and Hamada and Tachibana. In such

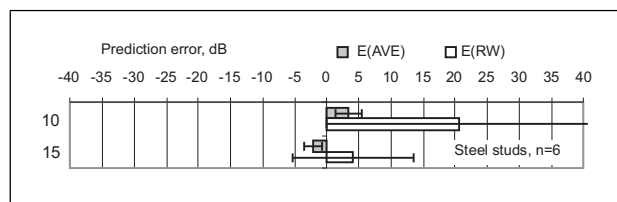


Figure 9. Average prediction error and standard deviation of  $E(R_w)$  and  $E(AVE)$  in category 4.

cases, the verification error cannot be explained by reading error. Errors in programming are not likely because the verification error reached zero at certain frequencies. If there were a significant error in programming, it would lead to a systematic error instead of a large standard deviation but a small average error. It is also possible that the authors have modified their calculation results towards measurement at such frequencies where the prediction error has been too large.

The smallest verification error was obtained with London's model. The reason for this is obvious: the model was the simplest to implement. The remaining verification error represents the reading error of the author. For some reason, the model of Beranek and Work could not be verified satisfactorily. There were large errors at specific frequencies, which increased the standard deviation. Fortunately, the models with the most serious verification errors were not among the most versatile and well-performing models.

## 5.2. Validation

In the following analysis, prediction error is defined as acceptable when the average error of  $E(R_w)$  or  $E(AVE)$  is below  $\pm 5$  dB, and the standard deviation is of the same order. Without cavity absorbent and studs (Figures 2 and 6), acceptable models regarding  $E(R_w)$  were those of London, Price and Crocker, Hamada and Tachibana, and Fahy 2001. Regarding  $E(AVE)$ , the acceptable models were those of London, Mulholland *et al.*, Cummings and Mulholland, Price and Crocker, Ookura and Saito, Fahy (1985), Au and Byrne, and Fahy (2001). Both tests were passed by the models of London, Price and Crocker, and Fahy (2001), which are recommended in category 1.

With full cavity absorption and without studs (Figures 3 and 7), acceptable models regarding  $E(R_w)$  were those of Sharp, Ookura and Saito, Heckl, Gu and Wang, and Au and Byrne. Regarding  $E(AVE)$ , acceptable models were those of Sharp, and Au and Byrne. Thus, the models of Sharp and Au and Byrne are recommended in category 2.

With rigid studs (Figures 4 and 8), five models were compared. Acceptable models regarding  $E(R_w)$  were those of Sharp and Davy. Regarding  $E(AVE)$ , acceptable models were those of Sharp, Qu and Wang, and Davy. The models of Sharp and Davy are recommended in category 3. However, neither of them was able to predict the dips in the low frequency region (200 Hz and 315 Hz in Figure 4). It was suggested in [7] that the dips are caused by sub-panel normal modes of the 2-mm thick steel. The

rigid studs of 600 mm stud division form very stiff boundary conditions, and sub-panels are formed between them. Their resonance frequency is naturally considerably higher (around 200 Hz) than the resonance panel calculated for the whole specimen size.

With flexible studs (Figures 5 and 8), only two models could be compared. The only acceptable model regarding  $E(R_w)$  was that of Davy. Regarding  $E(AVE)$ , both models were acceptable. Thus, the only recommended model in category 4 is Davy's.

At low frequencies, the variations between models are smaller than at high frequencies because the most important parameter is the total mass of the wall, which most models can deal with. However, at the mass-air-mass frequency, the variations are large. With double walls, the region of the mass-air-mass resonance is usually the most important frequency range for double walls regarding RW. Most of the models overestimate the  $SRI$  at high frequencies but this does not affect the accuracy of RW, which is mostly dependent on low frequency performance.

With an empty cavity (Figure 2), there are strong variations in the whole frequency region. With a sound-absorbing cavity (Figure 3), the variations are smaller. The models were different in how the cavity absorption was modelled. Some models expect that the cavity acts as a "dead" layer of specific acoustic impedance perpendicular to the panel. The impedance depends on material properties, such as thickness of absorption material or flow resistivity. Sound propagation in the plane of panels is omitted. Alternatively, some models use the absorption coefficient of the cavity boundaries, either the edges of the cavity or the inner surfaces of the panel. The author recommends the models having the former approach because the behaviour of sound in the cavity can be best understood according to Price and Crocker [15]. When the sound is trapped in the cavity, the cavity acts like a room where horizontal and vertical modes dominate at low frequencies, and perpendicular modes at high frequencies ( $\lambda < d/2$ ). The absorption of the inner surfaces of panels comes to play only at high frequencies. The problem is how to define the absorption coefficient of the cavity at different frequency ranges defined by the modal behaviour of the cavity. The most problematic situation that none of the present models can deal with is very thin and empty cavities ( $d < 30$  mm). In this case, the surface absorption of the panels can have a strong effect even at low frequencies because the in-plane sound fields cannot propagate freely in the cavity due to wall friction. It is extremely difficult to determine the effective cavity absorption in such cases. The application of nominal panel and edge absorption coefficients leads to strong underestimation of  $SRI$ . The behaviour of sound in the cavity will be studied in the future.

### 5.3. General discussion

This study gives evidence that one should not draw decisive conclusions about the accuracy of the prediction models on the basis of the validation in the original papers. Typically, the validation in the original papers has

been done for only one or a couple of experiments on real wall structures. It is also probable that the structures for the original papers were chosen so that a good fit would be obtained. That is, the original papers may contain partially false information misleading the gullible reader. It was also surprising that the prediction models presented in well-known textbooks do not work as well as expected, e.g. [26, 27, 30, 39]. The authors of future textbooks are encouraged to explain the limiting conditions of the presented models more carefully, and to include calculation examples for different wall types against which the model can be verified by the readers.

There were only two models that were able to predict the effect of flexible studs. According to Figures 5 and 9, only Davy's model gave satisfactory results. Gu and Wang's model resulted in strong overestimation of  $SRI$  at high frequencies, even though the prediction error of  $R_w$  was not large.

It is worth noting that Davy's model was the only one present in all four categories. However, its average performance with unstudded double walls was not satisfactory. Thus, the development of simple analytic and versatile prediction models has not yet finished.

There is an obvious need for a model that could deal with all types of double walls including empty, partially absorbing and fully absorbing cavities, and a studded cavity where the dynamic stiffness of the stud and stud division can be modified. The author has already published a preliminary version of such a model in Finnish [42] but it still needs some refinements regarding the effect of flexible studs and couplings. Recently, evidence has emerged that the method of attaching the panels to the studs is of great importance. The density of screws, screwing momentum, amount of glue, etc. should also be considered. According to Hongisto *et al.* [7] sub-panel resonances in densely studded double walls, especially in the case of rigid wood studs, greatly reduce the  $SRI$  at low frequencies. This effect is not considered in any of the present models. Also crossing double studs (vertical and horizontal studs overlapping) are difficult to model. How should the dynamic stiffness of the double stud be modelled if the dynamic stiffnesses of both individual studs are known? Could it be simplified by a point-type stud (crossing point) or does it still need a line-type stud but with a reduced mobility? Finally, there is not yet a simple model that considers the location and amount of the cavity absorbent although the effectiveness of using partial cavity absorption instead of full cavity absorption has been known 50 years already. It should also be kept in mind that real product development of lightweight multilayer walls presupposes simple prediction models having an accuracy of  $\pm 2 \dots 4$  dB in  $R_w$ . That is, the predictions do not need to be absolutely accurate at high frequencies because the  $R_w$  of lightweight structures mostly suffer from poor performance in middle and low frequency regions. Such models should be developed, which essentially improve accuracy in this region.

While preparing this study, it was necessary to expect that the measurements of [7] represent the truth and the

comparison of the prediction and the measurement data of [7] gives the true prediction error of the model. Otherwise, validation of prediction models would have been impossible. However, there is a risk that if the measurement results are unreliable, the validation process is ruined leading to false prediction error. The reliability means here that the measurement results differ from that of other laboratories. In this study, moreover, failed validation can lead to incorrect ranking between prediction models. There are two major sources of measurement uncertainties that could affect the validation process, namely inappropriate mounting and methodical errors.

There are two standardized methods to determine the *SRI*, pressure method ISO 140-3 [43] and intensity method ISO 15186-1 [44]. There is strong evidence that the *SRI* at low frequencies is slightly overestimated with ISO 140-3 [5]. The sound intensity method for low frequencies, ISO 15186-3, also has smaller inter-laboratory differences below 200 Hz [45]. The intensity method agrees better with mass law at low frequencies than the pressure method. Above 500 Hz, the differences between the pressure and intensity method are less than 1 dB. In [7], the pressure method was mainly used but it was found that the differences between the pressure and the intensity method were small, below 2 dB. The intensity method was used instead of the pressure method when reasonably high values of *SRI* were obtained and the pressure method gave underestimated values of *SRI*. Hongisto *et al.* have shown that, for structures having high sound insulation, intensity method is recommended instead of the pressure method to avoid the contribution of flanking transmission [46].

Inappropriate mounting of a double wall can lead to several problems, like flanking in aperture frame material, sound leaks due to improper sealing, and niche-effect due to inappropriate location of specimen in the test aperture [5]. Fausti *et al.* [47] reported an extensive round robin study including a double wall as a test sample. The reproducibility of *SRI* was considerably higher above 315 Hz than expected by ISO 140-2.  $R_w$  varied between 47 and 52 dB for a double wall structure. For single panels the results were between 26 and 29 dB which is within the limits of ISO 140-2 [48]. After this study, the preferred sill-to-reveal ratio was standardized [49].

The measurement data of [7] was obtained from an accredited test laboratory having a full conformance with ISO 140-3. Flanking was sufficiently suppressed and controlled, loudspeaker positions were properly selected, and room volumes were sufficiently large (81 and 113 m<sup>3</sup>). The intensity method was used in the case of high *SRI* values. The laboratory participated in a round robin test at the same year with good results [43]. The measurement data of double walls used in validation was obtained in highly controlled conditions where the niche-to-niche ratio was 1:2, specimens were properly sealed and all specimens were mounted by the same person.

There are two important factors that can reduce the measured *SRI* of double walls, namely flanking via frames of the test opening and edge transmission. Flanking via the

frames of the test opening can be strong especially with unstudded walls when intrinsic sound insulation is high. However, it is expected that such errors were avoided by the use of the intensity method.

Edge transmission can be relevant with flexible studs. Edge transmission means that peripherals of the wall have higher transmission than the middle part of the wall. The top and bottom rails are typically much stiffer than flexible studs in the middle part of the wall. In addition, the peripheral vertical studs are attached stiffly against the test frames which prevent the acoustic flexibility. However, the influence of edge transmission remains unknown. None of present prediction models is able to consider the edge transmission so that the topic remains for future studies.

## 6. Conclusions

Practical product development of sound insulation definitely benefits from simple, flexible, fast and accurate prediction models, which can be used in connection with carefully performed and documented laboratory tests. Seventeen well-known and analytic prediction models for airborne sound insulation were compared among each other and against measurement results. The variation between them was unacceptably high, more than 20 dB. Only few models could predict the sound insulation with acceptable accuracy, but none of them was capable of predicting the whole spread of commercial wall types consisting of varying layers of (flexible) studs, absorbents, cavities and partition panels. The need for better prediction models is obvious.

It can be said that no prediction model should be applied to any structure before it is verified against the original paper and validated against a structure which is as similar as possible to the structure to be modelled.

## Appendix

### Prediction models in chronological order

#### A1. Beranek and Work [8]

Structure IV:

$$\frac{p_0}{p_s} = x_1 \cosh(\Gamma_c d) + x_2 \sinh(\Gamma_c d),$$

$$x_1 = A + j \left( B + \frac{A \omega m'_2}{Z_0} \right),$$

$$x_2 = \frac{A Z_c}{Z_0} - \frac{B \omega m'_2}{Z_c} + j \frac{B Z_0}{Z_c},$$

$$A = \cos \frac{\omega l}{c_0} - \frac{\omega m'_1}{Z_0} \sin \frac{\omega l}{c_0},$$

$$B = \sin \frac{\omega l}{c_0} - \frac{\omega m'_1}{Z_0} \cos \frac{\omega l}{c_0},$$

where  $d$  is the thickness of cavity (m),  $\Gamma_c$  is the complex propagation factor of the cavity material,  $m'_1$  and  $m'_2$  are the surface masses of panels 1 and 2 (kg/m<sup>2</sup>),  $Z_0$  is the characteristic impedance of air,  $\rho_0 c_0$ , (kg/sm<sup>2</sup>),  $Z_c$  the

complex characteristic impedance of the cavity material,  $\omega$  is the angular frequency,  $l$  is the thickness of absorbent (m),  $c_0$  the speed of sound in air (343 m/s). Mulholland *et al.* [12] presented useful extension, which applies for an empty cavity:

$$\begin{aligned}\tau &= \left| \frac{p_t}{p_i} \right|^2 \\ &= \left| \frac{2Z_0 \coth(jk_0d) + \Phi}{Z_0 \coth(jk_0d) + \Phi + j\omega m' + Z_0} \cdot \frac{Z_0 \cosh \Phi}{\cosh(jk_0d) + \Phi(Z_0 + j\omega m')} \right|^2, \\ \Phi &= \coth^{-1}(1 - j\omega m'/Z_0),\end{aligned}$$

where  $k_0$  the wavenumber of sound in air (1/m).

## A2. London 1950 [9]

Method I (Real equations):

$$\begin{aligned}\tau(\theta) &= 1 + 4[R(R+1) + p^2v^2] \\ &\quad + 4\sin^2(k_0dv) \{ [R(R+1) + p^2v^2]^2 - p^2v^2 \} \\ &\quad - 4pv\sin(2k_0dv) [R(R+1) + p^2v^2], \\ v &= \cos \theta, \\ p &= a \left[ 1 - \left( \frac{f}{f_{cr}} \right)^2 (1 - q^2)^2 \right]; \quad a = \frac{\omega m'}{2Z_0}, \\ f_{cr} &= \frac{c_0^2}{2\pi} \sqrt{\frac{m'}{D}}, \quad D = \frac{Eh^3}{12(1 - \nu^2)},\end{aligned}$$

where  $R$  is London's panel resistance (=2.16 in this study),  $f_{cr}$  is the critical frequency of the panel (Hz),  $D$  is the bending stiffness of panel per unit width (Nm<sup>2</sup>),  $E$  is the Young's modulus of panel (N/m),  $h$  is the thickness of panel (m), and  $\nu$  is Poisson's ratio of panel.

Alternative method II (Integral equations):

$$\begin{aligned}\tau(\theta) &= \left| \frac{1}{1 + 2y + y^2[1 - e^{-2jk_0d \cos \theta}]} \right|^2, \\ y &= \frac{Z_p \cos \theta}{2Z_0}, \\ Z_p &= \frac{2R}{\cos \theta} + j\omega m' \left( 1 - \frac{f^2 \sin^4 \theta}{f_{cr}^2} \right).\end{aligned}$$

## A3. Mulholland, Parbrook and Cummings [12]

$$\begin{aligned}\tau(\theta) &= \left| \frac{x^2}{1 - (1 - \alpha_p)(1 - x)^2 e^{-2jk_0d \cos \theta}} \right|^2, \\ x &= \frac{1}{1 + j\omega m' \cos \theta / 2Z_0},\end{aligned}$$

where  $\alpha_p$  is the normal incidence absorption coefficient of the panel surfaces in the cavity.

## A4. Cummings and Mulholland [13]

$$\begin{aligned}\tau(\theta) &= \left| \frac{x_1 x_2}{1 - y} - \frac{x_1 x_2 y (1 - y^n)}{n(1 - y)^2} \right. \\ &\quad \cdot \left. \left[ 1 - \frac{\sqrt{1 - \alpha_c}(1 - y^n)}{1 - \sqrt{1 - \alpha_c} y^n} \right] \right|^2 \\ y &= (1 - x_1)(1 - x_2) e^{-2jk_0d \cos \theta}, \\ n(\theta) &= \frac{L_{a,c} \cdot L_{b,c}}{4d \tan \theta}, \\ x_i &= \frac{1}{1 + j\omega m'_i \cos \theta / 2\rho_0 c_0}, \quad i = 1, 2,\end{aligned}$$

where  $\alpha_c$  is the normal incidence absorption coefficient of the edges of the cavity,  $L_{a,c}$  is the height of cavity (m), and  $L_{b,c}$  is the width of cavity. Notes: The value of  $n$  will be extremely high with small  $\theta$ . Therefore,  $n$  was limited to 500 to avoid the collapse of calculation. The value of  $\alpha_c$  was limited to 0.99 for the same reason.

## A5. Price and Crocker [15]

Indices: 1: source room, 2: first panel, 3: cavity, 4: second panel, 5: receiving room.

$$SRI = 10 \log \frac{E_1}{E_5} - 10 \log \frac{V_1}{V_5} + 10 \log \frac{Sc_0 T_5}{24V_5 \ln 10},$$

where  $E$  is energy,  $V$  is room volume (m<sup>3</sup>), and  $T$  is reverberation time of the room(s).

$$\begin{aligned}\frac{E_1}{E_5} &= \left\{ \left( \eta_{5t} + \frac{\eta_{54}\eta_{35}}{\eta_{34}} \right) \left[ \frac{\eta_{4t}}{\eta_{34}} \left( \eta_{3t} - \frac{\eta_{23}\eta_{32}}{\eta_{2t}} \right) - \eta_{43} \right] \right. \\ &\quad \left. - \left( \eta_{45} + \frac{\eta_{4t}\eta_{35}}{\eta_{34}} \right) \left[ \frac{\eta_{54}}{\eta_{34}} \left( \eta_{3t} - \frac{\eta_{32}\eta_{23}}{\eta_{2t}} \right) - \eta_{53} \right] \right\} \\ &\quad \cdot \left\{ \left( \eta_{45} + \frac{\eta_{4t}\eta_{35}}{\eta_{34}} \right) \left( \eta_{13} + \frac{\eta_{12}\eta_{23}}{\eta_{2t}} \right) \right\}^{-1},\end{aligned}$$

where  $\eta_i$  is the loss factor of system  $i$ , and  $\eta_{ij}$  is the coupling loss factor from system  $i$  to system  $j$ .

$$\begin{aligned}\eta_{2t} &= \eta_2 + \eta_{21} + \eta_{23}, \\ \eta_{3t} &= \eta_3 + \eta_{31} + \eta_{32} + \eta_{34} + \eta_{35}, \\ \eta_{4t} &= \eta_4 + \eta_{43} + \eta_{45}, \\ \eta_{5t} &= \eta_5 + \eta_{53} + \eta_{54}, \\ n_i &= \frac{\sqrt{3}S_i}{2\pi h_i c_{L,i}} \quad (i = 2, 4), \\ n_i &= \frac{V_i \omega^2}{2\pi^2 c_0^3} \quad (i = 1, 5),\end{aligned}$$

where  $n_i$  is the modal density of the system  $i$  (s) and  $S$  is the area of panel.

$$\eta_{21} = \frac{\sigma_2}{\omega m_2}, \quad \eta_{45} = \frac{\sigma_4}{\omega m_4},$$

where  $m$  is the total mass of system  $i$  (kg) and  $\sigma$  is the radiation factor. Reciprocity principle says that  $\eta_{ij}n_i = \eta_{ji}n_j$ .

$$\eta_{23} = \begin{cases} \frac{2\sigma_2}{\omega m_2}, & f < f_{cr,2} \\ \frac{\sigma_2}{\omega m_2}, & f > f_{cr,2} \end{cases},$$

$$\eta_{43} = \begin{cases} \frac{2\sigma_4}{\omega m_4}, & f < f_{cr,4} \\ \frac{\sigma_4}{\omega m_4}, & f > f_{cr,4} \end{cases},$$

$$\sigma = S\rho_0 c_0 \begin{cases} \frac{2\lambda_{cr}\lambda_0 f}{S f_{cr}} g_1 + \frac{P\lambda_{cr}}{S} g_2, & f < f_{cr} \\ \sqrt{\frac{L_{a,c}}{\lambda_{cr}}} + \sqrt{\frac{L_{b,c}}{\lambda_{cr}}}, & f = f_{cr} \\ 1/\sqrt{1 - f_{cr}/f}, & f > f_{cr} \end{cases},$$

where  $P$  is the peripheral length of the panel.

$$g_1 = \begin{cases} \frac{4(1 - 2\kappa^2)}{\pi^4 \kappa \sqrt{1 - \kappa^2}}, & f < f_{cr}/2 \\ 0, & f > f_{cr}/2 \end{cases},$$

$$g_2 = \begin{cases} \frac{(1 - \kappa^2) \ln \frac{1+\kappa}{1-\kappa} + 2\kappa}{(2\pi)^2 \sqrt{(1 - \kappa^2)^3}}, & \kappa = \sqrt{f/f_{cr}} \end{cases},$$

$$10 \log \eta_{13} = -SRI_{rand,2} + 10 \log \frac{S c_0}{4V_1 \omega},$$

$$10 \log \eta_{53} = -SRI_{rand,4} + 10 \log \frac{S c_0}{4V_5 \omega}.$$

The random incidence mass law is calculated as follows [50, 51],

$$SRI_{rand} = SRI_0 - 10 \log(0.23 SRI_0) \text{ dB},$$

$$SRI_0 = 20 \log(m'_i f) - 42 \text{ dB},$$

where  $SRI_0$  is the normal incidence mass law.

$$n_3 = \begin{cases} \frac{S_c \omega}{2\pi c_0^2}, & f < f_d \\ \frac{V_3 \omega^2}{2\pi^2 c_0^3}, & f > f_d \end{cases}, \quad f_d = \frac{c_0}{2d}$$

$$\eta_3 = \begin{cases} \frac{S_{c,a} c_0 \alpha_c}{4V_3 \omega}, & f < f_d \\ \frac{S_{c,a} c_0 \alpha_c}{6V_3 \omega}, & f > f_d \end{cases},$$

where  $S_c$  is the area of the cavity ( $m^2$ ), and  $S_{c,a}$  is the area of the absorbent in the boundaries of the cavity,

$$S_c = L_{a,c} L_{b,c}, \quad S_{c,a} = 2d(L_{a,c} + L_{b,c}).$$

#### A6. Donato [17]

At  $f < 0.6f_{cr}$  and  $D \neq 1$ :

$$\tau(\theta) = \frac{2D^2}{D^2 + 4} \left[ \frac{1}{2A^2} \left[ \ln A^2 - \ln \left( \frac{(A - D)^2}{D^2 + 1} + 1 \right) \right] \right. \\ \left. - \frac{2}{A^2 D} \tan^{-1} A - \frac{1}{A^2} \left( D - \frac{2(D^2 + 1)}{D} \right) \right. \\ \left. \cdot \left( \tan^{-1}(A - D) + \tan^{-1} D \right) \right],$$

$$A = \frac{\omega m'}{Z_0}, \quad D = 2c_0/\omega d.$$

At  $f < 0.6f_{cr}$  and  $D \approx 1$ :

$$\tau = 4 \left( \frac{Z_0}{\omega m'} \right)^2.$$

At  $0.7f_{cr} > f > f_{cr}$  the surface mass  $m'$  is replaced by

$$m' \rightarrow m' \left[ 1 - \left( \frac{f}{f_{cr}} \right)^2 \sin^4 \theta \right].$$

At  $f = f_{cr}$ :

$$\tau(\theta) = \frac{8}{5.8} \left( \frac{\omega d}{c_0} \right)^{-2} \sqrt{\frac{\pi}{4.2}} \cdot 10^{-2} \simeq 1379(\omega d)^{-2}.$$

At  $f > f_{cr}$ :

$$\tau = \frac{2\pi \sqrt{1 - \frac{f_{cr}}{f}} \frac{Z_0 f}{\omega m' f_{cr}}}{1 + \frac{\omega m'}{Z_0} \eta \sqrt{1 - \frac{f_{cr}}{f}}}.$$

A correction to consider the effect of the finite size of the panel was also presented. The original equations were very complex. An exponential fitting procedure with 8-points was used to obtain an approximation of equation which is easier to solve numerically. It is added to the  $SRI$  prediction

$$\Delta SRI = 1.66 + e^{-(X-1)/0.89} \text{ dB}, \quad X = \frac{\omega L}{4.6c_0},$$

where  $L$  is the largest dimension of the panel.

#### A7. Sharp [32, 33]

The  $SRI$  of single panels are calculated by:

$$SRI = \begin{cases} 20 \log(m' f) - 48, & f < 0.5f_{cr}, \\ 20 \log(\omega m' / 2Z_0) \\ + 10 \log(2\eta\omega / \pi\omega_{cr}), & f > 0.5f_{cr}. \end{cases}$$

The frequency range  $0.5f_{cr} \dots f_{cr}$  is interpolated.

Double panel without studs are calculated by:

$$SRI = \begin{cases} SRI_m, & f < f_{mam}, \\ SRI_1 + SRI_2 + 20 \log(fd) - 29, & f_{mam} < f < f_l, \\ SRI_1 + SRI_2 + 6, & f > f_l, \end{cases}$$

$$SRI_{m'} = 20 \log(m'_i f) - 48 \text{ dB},$$

where  $m'_i$  is the total surface mass of the specimen:

$$m'_{tot} = m_1 + m_2.$$

The mass-air-mass resonance and the limit frequency are calculated by

$$f_{mam} = \frac{1}{2\pi} \sqrt{\frac{1.8\rho_0 c_0^2}{d} \frac{m'_1 + m'_2}{m'_1 m'_2}}, \quad f_l = \frac{c_0}{6d}.$$

The bridge frequency for rigid line connections is:

$$f_B = f_{mam} \sqrt{\frac{\pi b f_{cr}}{2c_0} \left( \frac{m'_1}{m'_1 + m'_2} \right)^2},$$

where  $b$  is the vertical or lateral distance of the interpanel connections (studs) in the cavity (m). Above the bridge frequency the sound reduction index cannot exceed the mass-law-curve  $SRI_{m'}$  more than by  $\Delta SRI_{m'}$ .

$$SRI = SRI_{m'} + \Delta SRI_{m'}, \quad f > f_B.$$

For rigid line connections:

$$\Delta SRI_{m'} = 10 \log(b f_{cr,h}) + 20 \log \frac{m'_1}{m'_1 + m'_2} - 18.$$

Here,  $f_{cr,h}$  is the higher critical frequency of the two panels and  $m'_1$  is the surface mass of the other panel (Note: This is an illogical selection rule, which leads to the failure of the model when the stiffness and mass of the two panels differ considerably from each other).

#### A8. Ookura and Saito [20]

Numbering of elements is: 1: panel in receiving room side, 2: cavity, 3: panel in source room side.

$$\tau(\theta) = \left| \frac{p_{11}}{p_i} \right|^2 = \left| \frac{p_{32}}{p_i} \right|^2 \cdot \left| \frac{p_{31} p_{21} p_{11}}{p_{32} p_{22} p_{12}} \right|^2,$$

$$\frac{p_{32}}{p_i} = \frac{p_i + p_r}{p_i} = \frac{2Z_{32}}{Z_{32} + Z_0 / \cos \theta},$$

where  $Z_{32}$  is the normal acoustic impedance of the panel in source room side,  $Z_{31}$  is the normal acoustic impedance on right hand side of the same panel. The left-hand impedance of this panel is calculated by the sum of the impedance at the right-hand side ( $Z_{31}$ ) and the relevant material ( $Z_3$ ):

$$Z_{32} = Z_{31} + Z_3 = Z_{22} + Z_3.$$

The surface impedance of the panel:

$$Z_i = \eta \omega m' \left( \frac{\omega}{\omega_{cr}} \right)^2 \sin^4 \theta + j \omega m' \left[ 1 - \left( \frac{\omega}{\omega_{cr}} \right)^2 \sin^4 \theta \right], \quad i = 1, 3.$$

The pressure ratio for panel (3):

$$\frac{p_{32}}{p_{31}} = \frac{Z_{32}}{Z_{31}} = \frac{Z_{31} + Z_3}{Z_{31}} = \frac{Z_{22} + Z_3}{Z_{22}}.$$

Cavity filled with absorbent:

$$\frac{p_{21}}{p_{21}} = \frac{\cosh(qd + \psi)}{\cosh \psi},$$

$$\psi = \coth^{-1} \frac{q Z_{21}}{\Gamma_c Z_c}, \quad \text{where } Z_{21} = Z_{12},$$

$$q = \Gamma_c \sqrt{1 + \left( \frac{k_0}{\Gamma_c} \right)^2 \sin^2 \theta},$$

$$Z_{22} = \frac{\Gamma_c Z_c}{q} \coth(qd + \psi).$$

Empty cavity:

$$\Gamma_c = j k_0, \quad Z_c = Z_0 = \rho_0 c_0,$$

$$\frac{p_{22}}{p_{21}} = \frac{\cosh(j k_0 d \cos \theta + \psi)}{\cosh \psi},$$

$$\psi = \coth^{-1} \frac{Z_{21} \cos \theta}{Z_0},$$

$$Z_{22} = \frac{Z_0}{\cos \theta} \coth(j k_0 d \cos \theta + \psi).$$

. In the boundary,  $Z_{21} = Z_{12}$ . The left-hand impedance of the panel (1) is the sum of the impedance at the right-hand side (receiving room side) and the panel impedance:

$$Z_{12} = Z_{11} + Z_1,$$

$$\frac{p_{12}}{p_{11}} = \frac{Z_{12}}{Z_{11}} = \frac{Z_{11} + Z_1}{Z_{11}}, \quad Z_{11} = \frac{Z_0}{\cos \theta}.$$

#### A9. Heckl [23]

$$\tau(\theta) = \left| 1 - \omega^2 \frac{m''_1 + m''_2}{2s'} + j \omega \frac{m''_1 + m''_2}{2Z} \right|^{-2} \cdot \left( 1 - \omega^2 \frac{m''_1 m''_2}{s'(m''_1 + m''_2)} + \frac{Z^2}{s'(m''_1 + m''_2)} \right)^{-2},$$

$$s' = \rho_0 c_0^2 / d,$$

where  $s'$  is the stiffness per unit area of the cavity filling ( $N/m^3$ ),

$$Z = Z_0 / \cos \theta,$$

$$m''_i = m'_i \left[ 1 - \left( \frac{k_0}{k_{Bi}} \sin \theta \right)^4 \right],$$

$$k_{Bi} = \sqrt[4]{\frac{\omega^2 M_i}{D'_i}}; \quad D'_i = D_i (1 + j \eta_i).$$

#### A10. Gu and Wang [36]

Flexible line connections, below  $f < 0.5 f_{cr}$ :

$$SRI = \begin{cases} 20 \log(m'_i f) - 48, & f < \frac{2}{3} f_{mam}, \\ SRI_1 + SRI_2 + 20 \log(f d) - 29, & f_{mam} < f < f_B, \\ SRI_1 + SRI_2 - 20 \log K'_B, & \\ -10 \log \frac{n_B l_B}{S f_{cr}} + 101, & f \geq f_B, \end{cases}$$

where  $k'_B$  is the lateral equivalent stiffness of a flexible stud per unit length ( $N/m$ ).

$$f_{mam} = \frac{1}{2\pi} \sqrt{\frac{(m'_1 + m'_2) \rho_0 c_0^2}{m'_1 m'_2 d}}.$$

The bridge frequency for line connections:

$$f_B = \frac{3.13 \cdot 10^6}{K'_B d} \sqrt{\frac{S f_{cr}}{n_B l_B}},$$

where  $n_B$  is the number of point or line connections in the wall. The lateral equivalent stiffness of the flexible stud per unit length:

$$K'_B = \frac{K_B}{l_B},$$

where  $l_B$  is the length of a line connection (m), and  $K_B$  is the lateral equivalent stiffness of a flexible stud (N). The lateral equivalent stiffness  $K_B$  is determined experimentally from the resonance frequency  $\omega_B$  that takes place when two masses  $m_1$  and  $m_2$  are connected to the flexible stud (spring)

$$\omega_B = \sqrt{\frac{m_1 + m_2}{m_1 m_2} K_B}.$$

The equations for point connections are not shown here.

#### A11. Fahy 1985 [26]

The model is restricted for sound-absorbing cavity and normal sound incidence angle.

$$\frac{\bar{p}_t}{\bar{p}_i} = - \left[ 4j\omega^3 \rho_0^2 c_0 \Gamma_c / \sinh(\Gamma_c d) \right] \cdot \left[ \Gamma_c^2 \bar{\alpha}_1 \bar{\alpha}_2 - \omega^2 \rho_0 \Gamma_c (\bar{\alpha}_1 + \bar{\alpha}_2) \coth(\Gamma_c d) + \omega_{mam}^4 \rho_0^2 \right]^{-1},$$

$$\bar{\alpha}_i = j\omega(\bar{z}_i + \rho_0 c_0),$$

$$\omega_{mam} = \sqrt{\frac{\rho_0 c_0^2}{d} \frac{m'_1 + m'_2}{m'_1 m'_2}},$$

$$\bar{z}_i = j\omega m'_i + r_i - j \frac{s_i}{\omega} = j\omega m'_i + \eta_i \omega_{0,i} m'_i - j \frac{\omega_{0,i}^2}{m'_i \omega}.$$

$r$ ,  $s$ , and  $\omega_0$  are the damping, the stiffness and the lowest natural frequency of the panel, respectively.

#### A12. Fahy 1985 [26]

The model is for empty cavity and diffuse sound incidence angle. Total transmission loss is the sum of two paths: air cavity and rigid studs.

$$\tau = \tau_{air} + \tau_{stud},$$

$$\tau_{air} = \left| \frac{\bar{p}_t}{\bar{p}_i} \right|^2 = \left| - \frac{2jZ_0^2 \sin(k_0 d \cos \theta) / \cos^2 \theta}{z'_1 z'_2 \sin^2(k_0 d \cos \theta) + Z_0^2 / \cos \theta} \right|^2,$$

$$\bar{z}'_i = \bar{z}_i + Z_0 [1 - j / \tan(k_0 d \cos \theta)] / \cos \theta,$$

$$\tau_{stud} = \frac{2n\rho_0^2 c_0 c_L}{\sqrt{3}\omega^3 \rho^2 h \cos \theta}, \quad c_L = \sqrt{\frac{E}{\rho(1-\nu^2)}},$$

where  $c_L$  is the velocity of the longitudinal wave of panel (m/s), and  $\rho$  is the density of panel (kg/m<sup>3</sup>). The transmission coefficient for studs can be applied to all models of Fahy.

#### A13. Hamada and Tachibana [24]

Each subsystem of the wall can be described as a cascade matrix (F-matrix). The matrix of the three-layer system is a multiplication of three matrixes:

$$F = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = F_1 \cdot F_2 \cdot F_3.$$

The transmission coefficient of double panel:

$$\tau(\theta) = \left| \frac{2}{A + B \frac{\cos \theta}{Z_0} + C \frac{Z_0}{\cos \theta} + D} \right|^2.$$

The F-matrix for a panel material:

$$F_p = \begin{bmatrix} 1 & Z_p \\ 0 & 1 \end{bmatrix},$$

$$Z_p = \frac{2R}{\cos \theta} + j\omega m' \left[ 1 - \left( \frac{f}{f_{cr}} \right)^2 \sin^4 \theta \right].$$

For sound-absorbing cavity:

$$F_c = \begin{bmatrix} \cosh(qd) & Z_{cx} \sinh(qd) \\ \sinh(qd)/Z_{cx} & \cosh(qd) \end{bmatrix},$$

$$q = \Gamma_c \sqrt{1 - \left( \frac{jk_0 \sin \theta}{\Gamma_c} \right)^2}, \quad Z_{cx} = \frac{\Gamma_c Z_c}{q}.$$

For empty cavity:  $\Gamma_c = jk_0$ ,  $Z_c = \rho_0 c_0$ .

$$F_c = \begin{bmatrix} \cos(k_0 d \cos \theta) & \frac{jZ_0 \sin(k_0 d \cos \theta)}{\cos \theta} \\ \frac{j \cos \theta \sin(k_0 d \cos \theta)}{Z_0} & \cos(k_0 d \cos \theta) \end{bmatrix}.$$

#### A14. Au and Byrne [28, 29]

The model has been presented also by Ver [27]. The input impedance  $Z_I$  of each material layer depends on the terminal impedance of that material layer  $Z_T$  and the specific impedance of the material in question. The calculation starts from the receiving room side. The input impedance of the receiving room:

$$Z_I = Z_0 / \cos \theta.$$

This term is the terminal impedance  $Z_T$  of the first panel. The impedance of second and first panel (layers 2 and 4 in model):

$$Z_2 = Z_1 + \frac{\eta_2 D_2 k_x^4}{\omega} + j \left( \omega m'_2 - \frac{D_2 k_x^4}{\omega} \right)$$

$$Z_4 = Z_3 + \frac{\eta_1 D_1 k_x^4}{\omega} + j \left( \omega m'_1 - \frac{D_1 k_x^4}{\omega} \right).$$

The impedance of porous layer, or air cavity:

$$Z_3 = Z_c \frac{\Gamma_c}{\Gamma_{cy}} \frac{\left( 1 + \frac{Z_c \Gamma_c}{Z_2 \Gamma_{cy}} \right) e^{j\Gamma_{cy} d} + \left( 1 - \frac{Z_c \Gamma_c}{Z_2 \Gamma_{cy}} \right) e^{-j\Gamma_{cy} d}}{\left( 1 + \frac{Z_c \Gamma_c}{Z_2 \Gamma_{cy}} \right) e^{j\Gamma_{cy} d} - \left( 1 - \frac{Z_c \Gamma_c}{Z_2 \Gamma_{cy}} \right) e^{-j\Gamma_{cy} d}}.$$

The propagation factor in y-dimension:

$$\begin{aligned}\Gamma_{cy}^2 &= \Gamma_c^2 - k_x^2, \quad k_x = k_0 \sin \theta, \\ Z_c &= Z_0 \sqrt{1 - j r' / \rho_0 \omega}, \\ \Gamma_c &= k_0 \sqrt{1 - j r' / \rho_0 \omega},\end{aligned}$$

where  $r'$  is the specific flow resistance (flow resistivity) of the cavity absorbent (Pa s/m<sup>2</sup>).

$$p_4 = p_i \frac{2A}{A+1}, \quad A = \frac{Z_4 \cos \theta}{Z_0}.$$

$p_4$  acts as the input pressure  $p_I$  for layer 3 (cavity). The transmitted pressure over an impervious panel is:

$$\begin{aligned}p_2 &= p_1 Z_2 / Z_1 && \text{second panel,} \\ p_4 &= p_3 Z_4 / Z_3 && \text{first panel.}\end{aligned}$$

The transmitted pressure over the cavity absorbent is calculated by

$$\begin{aligned}p_2 &= \frac{p_3}{2} \left[ \left( 1 + \frac{Z_c \Gamma_c}{Z_3 \Gamma_{cy}} \right) e^{-j \Gamma_{cy} d} + \left( 1 - \frac{Z_c \Gamma_c}{Z_3 \Gamma_{cy}} \right) e^{j \Gamma_{cy} d} \right], \\ \tau(\theta) &= \left| \frac{p_1}{p_i} \right|^2 = \frac{Z_1 Z_3}{Z_2 Z_4} \frac{A}{A+1} \left[ \left( 1 + \frac{Z_c \Gamma_c}{Z_3 \Gamma_{cy}} \right) e^{-j \Gamma_{cy} d} \right. \\ &\quad \left. + \left( 1 - \frac{Z_c \Gamma_c}{Z_3 \Gamma_{cy}} \right) e^{j \Gamma_{cy} d} \right].\end{aligned}$$

#### A15. Davy [44]

The total sound transmission coefficient is the sum of airborne and stud-borne paths:

$$\tau = \tau_a + \tau_s,$$

$\omega < \omega_{mam}$ :

$$\begin{aligned}\tau &= \frac{1}{a^2} \ln \frac{1 + a^2}{1 + a^2 \lambda / (2\pi \sqrt{S})}, \\ a &= \frac{\omega(m'_1 + m'_2)}{2\rho_0 c_0},\end{aligned}$$

where  $S$  is the wall area. Above  $\omega_{mam}$  and below  $f_{c1}$  the airborne sound transmission coefficient,  $\tau_a$ , and the stud-borne sound transmission coefficient,  $\tau_s$ , are calculated separately.

$\omega_{mam} < \omega < \omega_{cr}$ :

$$\begin{aligned}\tau_a &= \left[ 1 - \lambda / (2\pi \sqrt{S}) \right] \cdot \left[ \left( \frac{m_1^2 + m_2^2}{2m_1 m_2} \right. \right. \\ &\quad \left. \left. + a_1 a_2 a_c \lambda / (2\pi \sqrt{S}) \right) \left( \frac{m_1^2 + m_2^2}{2m_1 m_2} + a_1 a_2 a_c \right) \right]^{-1}, \\ a_i &= \frac{\omega m'_i}{2\rho_0 c_0} [1 - \kappa_i], \quad \kappa_i = \sqrt{\omega / \omega_{cr}}.\end{aligned}$$

$\omega > \omega_{cr}$ :

$$\begin{aligned}\tau_a &= \frac{\pi(\kappa_1 + \kappa_2)n}{4B_1^2 B_2^2 \eta_1 \eta_2 \kappa_1 \kappa_2 (n^2 + \nu^2) a_c^2} \\ B_i &= \frac{\omega m'_i}{2\rho_0 c_0}, \quad n = \eta_1 \kappa_2 + \eta_2 \kappa_1, \quad \nu = 4(\eta_1 - \eta_2).\end{aligned}$$

The stud-borne path ( $\omega > \omega_{mam}$ ):

$$\begin{aligned}\tau_s &= \frac{64\rho_0^2 c_0^3 F}{[g^2 + (\omega^{1.5} m'_1 m'_2 c_0 C_M - g)^2] \beta b \omega^2}, \\ g &= m'_1 \sqrt{\omega_{cr,2}} + m'_2 \sqrt{\omega_{cr,1}}, \\ F &= \begin{cases} 2, & \omega < \omega_{cr,1} \text{ and } \omega < \omega_{cr,2}, \\ \frac{\pi \omega_{cr,1} \omega_{cr,2}}{16\omega^2 \eta_1 \eta_2}, & \text{elsewhere,} \end{cases} \\ \beta &= (1 - \kappa_1^2)^2 (1 - \kappa_2^2)^2.\end{aligned}$$

The mechanical compliance of the studs  $C_M$  (m<sup>2</sup>/N) is the reciprocal of the mechanical stiffness of the stud,  $K$ .

The corrections presented in [45] to substitute the empirical factor of 2 in the equation of  $F$  have not been taken into account due to added complexity.

#### A16. Kang 2000 [31]

$$\begin{aligned}\tau(\theta) &= \left| \left[ \frac{-2j(Z_0 \sec \theta)^2}{k_0 d \cos \theta} \right] \left[ \left( Z_p + \frac{Z_0}{\cos \theta} - \frac{j Z_0}{k_0 d \cos \theta} \right)^2 \right. \right. \\ &\quad \left. \left. + \left( \frac{Z_0}{k_0 d \cos^2 \theta} \right)^2 \right]^{-1} \right|, \\ \tau_d &= \frac{\int_0^{\pi/2} G(\theta) \tau(\theta) \sin \theta \cos \theta d\theta}{\int_0^{\pi/2} G(\theta) \sin \theta \cos \theta d\theta}, \\ G(\theta) &= e^{-\beta \theta^2}, \\ Z_p &= j\omega m' - j \frac{\omega^3 D \sin^4 \theta}{c_0^4}, \quad D = D(1 - j\eta).\end{aligned}$$

Note:  $\theta_i = 90^\circ$ . It was recommended that  $\beta = 2$ .

#### A17. Fahy 2001 [27]

The model is for sound-absorbing cavity and normal sound incidence angle.

$$\begin{aligned}\frac{p_t}{p_i} &= 2z'_c \left[ j[z'_2 z'_1 + z'_2 + z'_1 + (z'_c)^2] \sin(k'd) \right. \\ &\quad \left. + z'_c [z'_2 + z'_1 + 2] \cos(k'd) \right]^{-1}, \\ k' &= \beta - j\alpha \approx \frac{\omega}{c_0} \sqrt{\frac{s}{K}} - j \frac{r' Y}{2\rho_0 c_0} \sqrt{\frac{s}{K}}, \\ z'_i &= z_i / \rho_0 c_0, \\ z_i &= j\omega m'_i + r_i - j s_i / \omega \\ &= j\omega m'_i + \eta_i \omega_{0,i} m'_i - j \omega_{0,i}^2 / (m'_i \omega), \\ z'_c &= z_c / \rho_0 c_0 \approx \sqrt{\frac{K s}{Y^2}} - j \frac{r'}{2\omega \rho_0} \sqrt{\frac{K}{s}},\end{aligned}$$

where  $Y$  is the the porosity of absorbent and  $s$  is the structure factor of absorbent. It was assumed that  $Y = s \approx 1$ ,  $K \approx 0.7 \dots 1$ .

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